DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer Regular Examination – 2023

Course: B. Tech. **Branch :CSE/Computer Engineering** **Semester : VI**

Subject Code & Name: BTCOC603 Machine Learning

Max Marks: 60 **Date:17/07/2023** **Duration: 3 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		12
A)	**Define machine learning. Explain its common areas of applications.**	(Remember)	6
B)	**Describe working of decision tree with suitable example.**	(Apply)	6
C)	**State the purpose of confusion metrics. Elaborate various terms associated with it.**	(Understand)	6
Q.2	**Solve Any Two of the following.**		12
A)	**State Bayestheorem. Demonstrateit with suitable example.**	(Apply)	6
B)	**With respect to SVM explain the terms: 1. Hyperplane 2. Maximum Margin Hyperplane (MMH) 3. Support Vector.**	(Understand)	6
C)	**Differentiate between linear and non-linear SVM.**	(Analyze)	6
Q. 3	**Solve Any Two of the following.**		12
A)	**Compare between Biological Neural Network (BNN) and Artificial Neural Network (ANN).**	(Analyze)	6
B)	**Explain reinforcement learning in Artificial Neural Network (ANN).**	(Understand)	6
C)	**Describe feedforward and feedback neural network with proper diagram.**	(Understand)	6
Q.4	**Solve Any Two of the following.**		12
A)	**Explain concept of VC dimensions with example.**	(Remember)	6
B)	**Discuss Ensembled Learning. State applications of ensembled learning.**	(Understand)	6
C)	**Describe terms: Bagging, Boosting and Stacking.**	(Understand)	6
Q. 5	**Solve Any Two of the following.**		12
A)	**Explain clustering with example. State need of clustering.**	(Understand)	6

B)	**State the working of K-means clustering algorithm.**	(Remember)	6
C)	**Differentiate between Agglomerative and Divisive hierarchical clustering.**	(Analyze)	6

***** End *****